

Superposition in Circuit Analysis

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1. Introduction

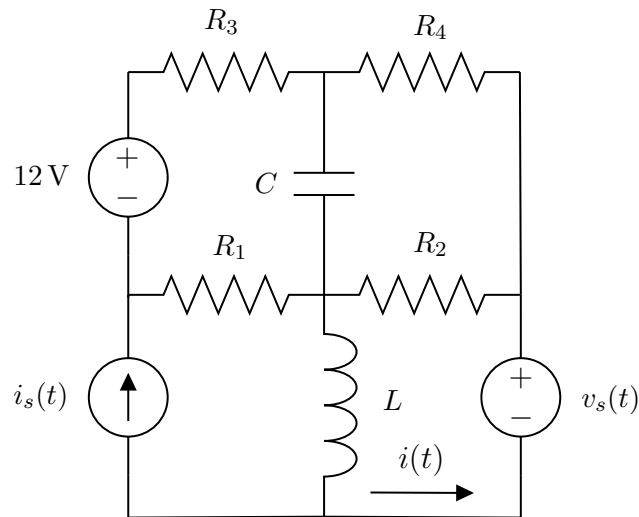
During my second year, the Superposition theorem in circuit analysis always fascinated me because it can be used to solve both linear and non-linear problems by considering the effect of each independent source separately. Therefore, I am writing this paper with a belief that it could be taught in a more systematic approach with more advanced problems.

We may already be familiar with the Superposition Theorem, as it appears in many areas of mechanics and electromagnetics. In circuit analysis, the principle states that, *for a linear circuit containing multiple independent sources, the voltage across or current through any circuit element is equal to the algebraic sum of the responses produced by each independent source acting individually.*

The most important requirement for applying the Superposition theorem is that the circuit must be linear. When using the theorem, the effect of each independent source is analyzed separately while all other independent sources are deactivated.

2. Applications

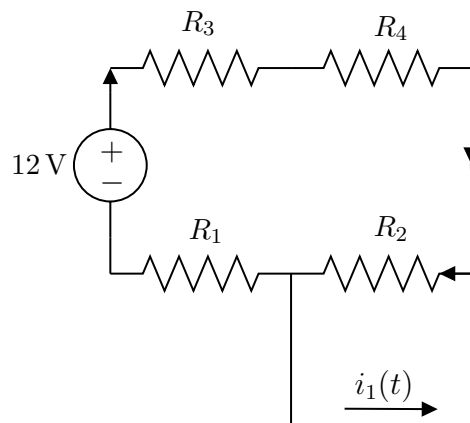
Given a circuit with the source $i_s(t) = 4 \cos(3t) A$ and $v_s(t) = 3 \sin(4t) V$, where the circuit parameters are $R_1 = R_2 = R_3 = R_4 = 1 \Omega$, $L = 1H$ and $C = \frac{1}{12} F$. Use Superposition theorem to solve for the current $i(t)$.



Solution. We first solve for DC signal. In DC steady state,

- We turn off other **current sources** so they become **open circuits** and turn off other **voltage sources** so they become **short circuits**
- Capacitors act like **open circuits**, inductors act like **short circuits**

Thus, the diagram is simplified as

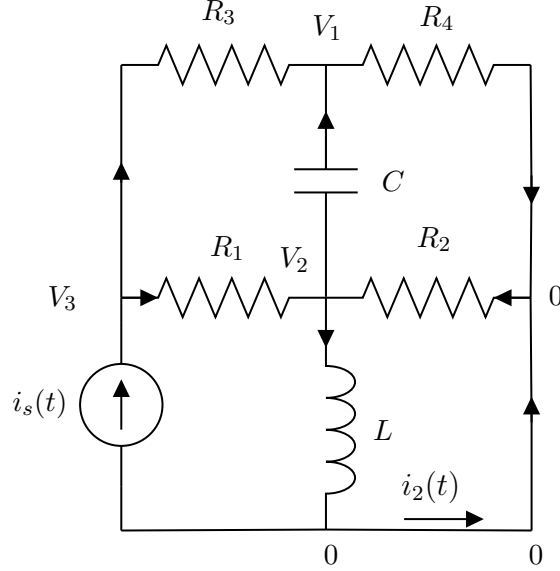


We can easily compute current $i(t)$ in DC steady state by using Ohm's law

$$i_1(t) = -\frac{12}{R_1 + R_3 + R_4} = -\frac{12}{1 + 1 + 1} = -4 \text{ A}$$

Noted that the minus sign indicates the current in this state is flowing in the opposite direction of the given problem.

Next, we solve the current in AC steady state due to the current source. The same principle applies here, we turn off other sources



According to Kirchhoff's Current Law,

$$\frac{V_2 - V_1}{1/(i\omega C)} + \frac{V_3 - V_1}{R_3} = \frac{V_1 - 0}{R_4} \quad (1)$$

$$\Rightarrow \frac{i}{4}(V_2 - V_1) + V_3 - V_1 = V_1$$

$$\frac{V_3 - V_2}{R_1} + \frac{0 - V_2}{R_2} = \frac{V_2 - V_1}{1/(i\omega C)} + \frac{V_2 - 0}{i\omega L} \quad (2)$$

$$\Rightarrow V_3 - V_2 - V_2 = \frac{i}{4}(V_2 - V_1) - \frac{i}{3}V_2$$

$$\frac{V_3 - V_2}{R_1} + \frac{V_3 - V_1}{R_3} = I_s \quad (3)$$

$$\Rightarrow V_3 - V_2 + V_3 - V_1 = 4 \angle 0$$

Solving this system of equations, we yield

$$V_1 = \frac{20 - 8i}{10 - 5i}, \quad V_2 = \frac{48 + 12i}{25}, \quad V_3 = \frac{98 + 8i}{25}.$$

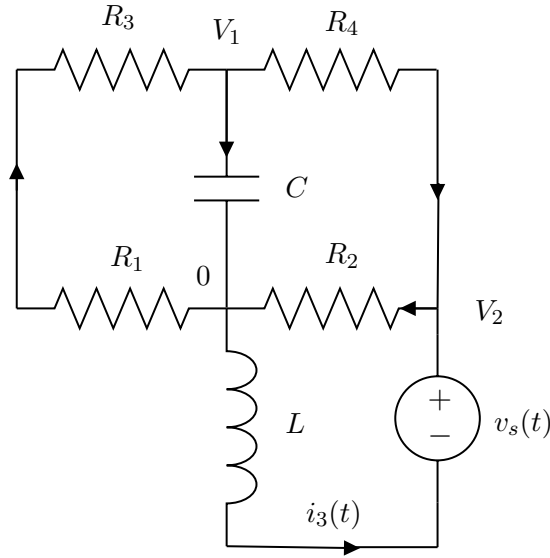
Therefore, we can solve for phasor I_2

$$I_2 = \frac{V_2 - 0}{i\omega L} - I_s = \frac{1}{3i} \frac{48 + 12i}{25} - 4\angle 0 = -3.84 - 0.64i$$

So we can convert it back to time domain

$$i_2(t) = \frac{16}{25} \sqrt{37} \cos \left(3t + \pi + \arctan \frac{1}{6} \right)$$

Lastly, we focus on the AC voltage source by turning off other sources,



Solving the last piece is just a repetitive work. According to Kirchhoff's Current Law,

$$\frac{0 - V_1}{R_1 + R_3} = \frac{V_1 - 0}{1/(i\omega C)} + \frac{V_1 - V_2}{R_4} \quad (4)$$

$$\Rightarrow -\frac{V_1}{2} = \frac{i}{3} V_1 + V_1 - V_2$$

$$\frac{V_2 - 0}{R_2} = \frac{V_1 - V_2}{R_4} + \frac{0 + V_s - V_2}{i\omega L} \quad (5)$$

$$\Rightarrow V_2 = V_1 - V_2 + \frac{3\angle 0 - V_2}{4i}$$

Solving this system of equations, we yield

$$V_1 = \frac{-8}{7 - 50i}, \quad V_2 = \frac{-27 - 6i}{7 - 50i}.$$

Hence, phasor I_3 can be calculated as

$$I_3 = \frac{0 + V_s - V_2}{i\omega L} = \frac{1}{4i} \left(3\angle 0 - \frac{-27 - 6i}{7 - 50i} \right) = \frac{1312}{10196} - \frac{10085}{10196}i.$$

We can convert it back to time domain

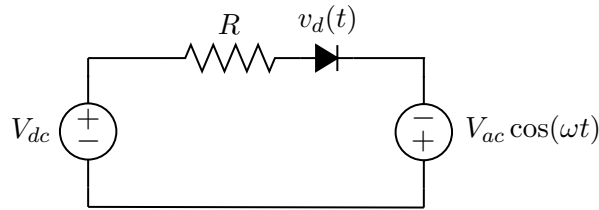
$$i_3(t) = \frac{12\sqrt{10}}{\sqrt{2549}} \sin \left(4t + \arctan \frac{157}{29} \right)$$

By superposition, we just need to add up three signals, then we are done!

$$i(t) = i_1(t) + i_2(t) + i_3(t) \approx -4 + 3.893 \cos(3t + 2.976) + 0.752 \sin(4t + 1.388)$$

Up until this point, the solution has been somewhat tedious...especially if the problem has more than 4 independent sources since we have to redraw the circuit over again and find the current for each source. In fact, we are still able to solve this problem using Laplace Transform to get the transient and steady state response at the same time (it is encouraged to work through this as an exercise). So what is the point of learning this tool? The next problem will have an answer to this question.

Problem 1.2. Given the circuit with $V_{ac} = 4$ V, $V_{dc} = 1$ V and $R = 3\Omega$. A diode exhibits a voltage of 0.7 V at 1.6 mA with $n = 2$ and $V_T = 0.025$ V. Calculate voltage of the real diode in the form of $v_d(t) = \alpha + \beta \cos(\omega t)$. Note: The circuit is in steady state.



Solution. Before tackling this problem using superposition, I would like to draw attention to the direct approach. According to Kirchhoff's Voltage Law, we can write

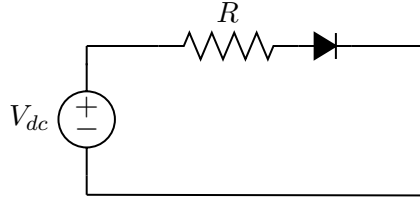
$$V_{dc} + V_{ac} \cos(\omega t) = v_d + i_d R$$

$$i_d = I_s (e^{v_d/(nV_T)} - 1)$$

It may seem that having two equations and two unknowns should be enough to solve the system directly. However, the situation is more complicated. The diode equation is nonlinear because of its exponential dependence on v_d . In general, such systems do not admit simple closed-form solutions and are usually solved using Lambert W functions or suitable approximations such as small-signal linearization. In this case, we will use superposition for the second approach.

However, you might think: “Superposition is only valid for linear systems.” That is true. This is precisely why we linearize the circuit around its operating point. After linearization, the circuit behaves approximately linearly for small AC variations, allowing the DC bias and AC signals to be analyzed separately using superposition.

In DC steady state, we can turn off the AC source. The diagram can be redrawn as



We first find the reverse saturation current I_s

$$I_s \approx \frac{I_d}{e^{V_D/(nV_T)}} = \frac{1.6 \times 10^{-3}}{e^{0.7/(2 \cdot 0.025)}} \approx 1.33 \times 10^{-9} \text{ A}$$

Since V_{dc} is significantly large, we apply Kirchhoff's Voltage Law

$$V_{dc} = \alpha + Ri_d$$

Then our goal is to solve the non-linear equation

$$\alpha + 3I_s(e^{\alpha/(2 \cdot 0.025)} - 1) = 1$$

$$\text{Let } f(\alpha) = 1 + 3I_s(e^{20\alpha} - 1) - 1 \Rightarrow f'(\alpha) = 1 + 60I_se^{20\alpha}$$

We can use Newton-Raphson method to approximate the solution

$$\alpha_{k+1} = \alpha_k - \frac{f(\alpha_k)}{f'(\alpha_k)} \quad \text{with} \quad \alpha_0 = 0.7 \text{ V}$$

Then the equation becomes

$$\alpha_{k+1} = \frac{1 + 3I_s(e^{20\alpha_k} - 1) - 1}{1 + 60I_se^{20\alpha_k}} \quad \text{with} \quad \alpha_0 = 0.7 \text{ V}$$

The first six iterates are

$$\begin{array}{lll} \alpha_1 = 0.9693 & \alpha_2 = 0.9230 & \alpha_3 = 0.8867 \\ \alpha_4 = 0.8692 & \alpha_5 = 0.8664 & \alpha_6 = 0.8663 \end{array}$$

As the number of iterations increases, the sequence converges to

$$\alpha \approx 0.8663 \text{ V}$$

Now we recall the Shockley diode equation

$$i \approx I_se^{v/(nV_T)} \Rightarrow 1 = \frac{I_s}{nV_T} e^{v/(nV_T)} \frac{dv}{di}$$

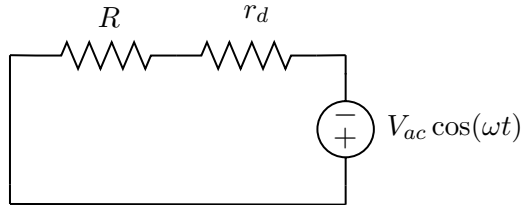
Rearranging two sides of the equation,

$$\frac{dv}{di} = \frac{nV_T}{I_se^{\alpha/(nV_T)}} = \frac{nV_T}{I_Q}$$

The derivative dv/di is called dynamic resistance of the diode. The subscript Q refers to the quiescent point, which represents the steady-state DC operating condition of an electronic device. Therefore,

$$r_d = \frac{nV_T}{I_Q} = \frac{2 \cdot 0.025}{1.33 \times 10^{-9} \cdot e^{0.866/(2 \cdot 0.025)}} \approx 1.12 \Omega$$

Now we find the voltage of the diode in AC steady state by replacing the diode with its dynamic resistor



Using voltage divider, we yield

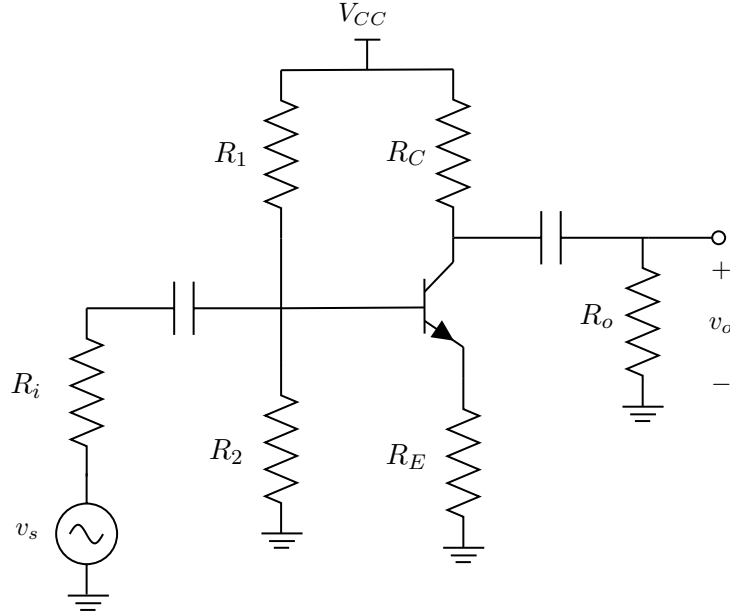
$$\beta = \frac{r_d}{r_d + R} V_{ac} \cos(\omega t) = \frac{1.12}{1.12 + 3} \cdot 4 \cos(\omega t) \approx 1.09 \cos(\omega t) \text{ V}$$

Therefore, by superposition

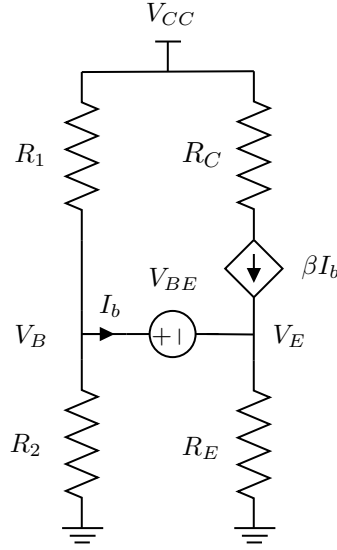
$$v_d(t) = 0.866 + 1.09 \cos(\omega t) \text{ V}$$

So far, we can safely conclude that the Superposition Principle is extremely useful for analyzing small-signal models of nonlinear devices such as diodes, BJTs, and MOSFETs. However, because these devices are inherently nonlinear, the principle becomes valid only after linearization around an operating point, and therefore applies only as an approximation for sufficiently small signals.

Problem 1.3. Given transistor in an amplifier below with $\beta = 100$, $V_{BE} = 0.7 \text{ V}$, and $V_{CE(sat)} = 0.2 \text{ V}$. Meanwhile, $V_{CC} = 5 \text{ V}$, $R_i = 50 \Omega$, $R_o = 110 \Omega$, $R_1 = 6 \text{ k}\Omega$, $R_2 = 4 \text{ k}\Omega$, $R_C = 800 \Omega$ and $R_E = 640 \Omega$. Assume $V_T = 0.025 \text{ V}$. If the input signal is $v_i(t) = 10 \sin(2t)$ determine the output voltage $v_o(t)$



Solution. For DC analysis, we turn off the AC source and the coupling capacitors become open circuits. Therefore, the circuit diagram is simplified to



You might ask why we should not use the more accurate non-linear model $I_C = I_s e^{V_{BE}/V_T}$ for this analysis. While this model is indeed more precise, it leads to much more complicated equations. Therefore, when the BJT operates in the active region, we typically use the simpler 0.7 V approximation for practical DC analysis and this is good enough. According to Kirchhoff's Current Law,

$$\frac{V_{CC} - V_B}{R_1} = I_B + \frac{V_B}{R_2} \quad (6)$$

$$\Rightarrow \frac{5 - V_B}{6000} = I_B + \frac{V_B}{4000}$$

$$(\beta + 1)I_B = \frac{V_E}{R_E} \quad (7)$$

$$\Rightarrow 101I_B = \frac{V_E}{640}$$

$$V_B - V_E = V_{BE} \quad (8)$$

$$\Rightarrow V_B - V_E = 0.7$$

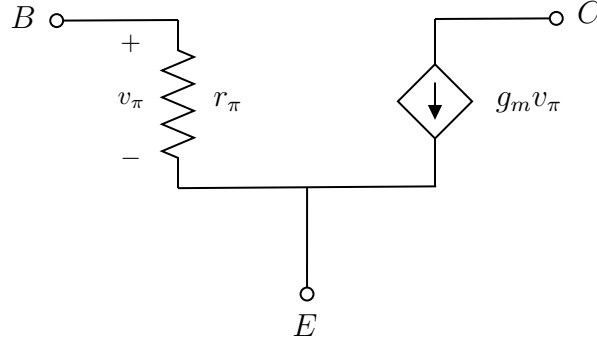
Solving this system of equations, we yield

$$V_B \approx 1.953 \text{ V}, \quad V_E = 1.253 \text{ V}, \quad I_B = 1.939 \times 10^{-5} \text{ A}$$

As a result, $V_C = V_{CC} - \beta I_B R_C = 5 - 100 \cdot 1.939 \times 10^{-5} \cdot 800 \approx 3.449 \text{ V}$.

Since $V_{CE} = V_C - V_E \approx 2.2 \text{ V} > V_{CE(sat)} = 0.2 \text{ V}$, the BJT is in active region and our assumption was correct.

For AC analysis, we short the constant voltage source and coupling capacitors become ideal wires. Additionally, we introduce the hybrid- π model, which is a small-signal equivalent circuit used to analyze a BJT transistor for AC signals.



Now we will use the non-linear BJT equation to derive two important parameters: the small-signal resistance seen looking into the base-emitter junction r_π and small-signal transconductance g_m .

The small-signal transconductance of a BJT is defined as

$$g_m = \frac{dI_c}{dV_{BE}}$$

This derivative can be found from the Shockley diode equation

$$I_C = I_s e^{V_{BE}/V_T}$$

Differentiating two sides of the equation with respect to V_{BE} ,

$$\frac{dI_c}{dV_{BE}} = \frac{I_s}{V_T} e^{V_{BE}/V_T} = \frac{I_{CQ}}{V_T}$$

Therefore,

$$g_m = \frac{I_{CQ}}{V_T}$$

Meanwhile, small-signal resistance seen looking into the base-emitter junction r_π is defined as

$$r_\pi = \frac{dV_{BE}}{dI_C} \frac{dI_C}{dI_B} = \frac{\beta}{g_m}$$

Since $g_m = I_{CQ}/V_T = \beta I_{BQ}/V_T$, we can express r_π as

$$r_\pi = \frac{\beta}{\beta I_{BQ}/V_T} = \frac{V_T}{I_{BQ}}$$

In summary,

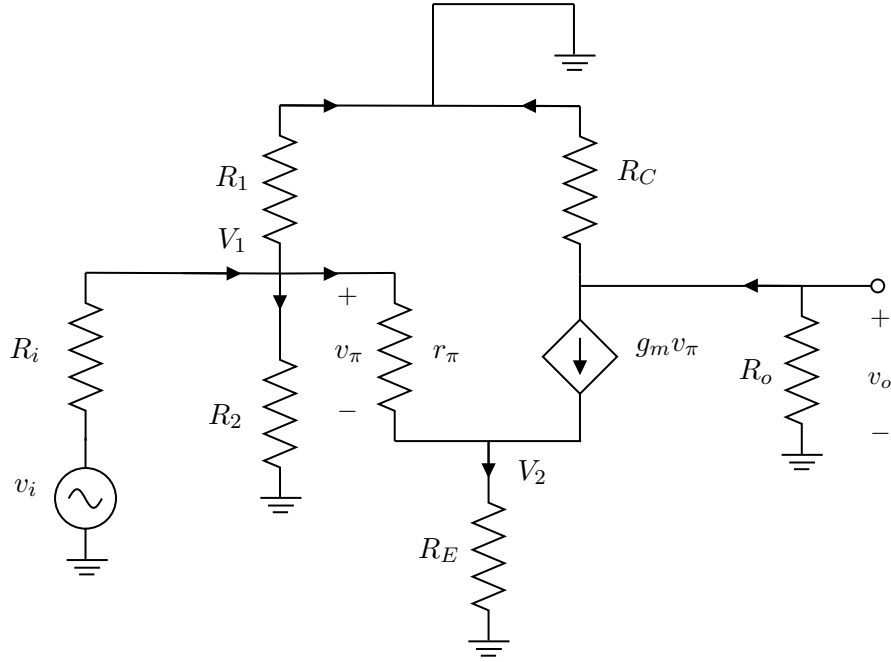
$$r_\pi = \frac{V_T}{I_{BQ}}, \quad g_m = \frac{\beta}{r_\pi}$$

Once again, I_{BQ} and I_{CQ} refer to the quiescent base and collector currents of the BJT. Based on the information from the question, we can calculate

$$r_\pi = \frac{0.025}{1.939 \times 10^{-5}} \approx 1289.23 \, \Omega$$

$$g_m = \frac{100}{1289.23} \approx 7.757 \times 10^{-2} \, \text{A/V}$$

Hence, the circuit diagram can be redrawn as



According to Kirchhoff's Current Law,

$$\frac{v_i - V_1}{R_i} = \frac{V_1 - 0}{R_1} + \frac{V_1 - 0}{R_2} + \frac{V_1 - V_3}{r_\pi} \quad (9)$$

$$\Rightarrow \frac{v_i - V_1}{50} = \frac{V_1 - 0}{6000} + \frac{V_1 - 0}{4000} + \frac{V_1 - V_2}{r_\pi}$$

$$\frac{V_2 - 0}{R_E} = \frac{V_1 - V_2}{r_\pi} + g_m v_\pi \quad (10)$$

$$\Rightarrow \frac{V_2}{640} = \frac{V_1 - V_2}{1289.23} + 7.757 \times 10^{-2} v_\pi$$

$$\frac{0 - v_o}{R_o} = \frac{v_o - 0}{R_C} + g_m v_\pi \quad (11)$$

$$\Rightarrow \frac{-v_o}{110} = \frac{v_o}{800} + 7.757 \times 10^{-2} v_\pi$$

$$v_\pi = v_1 - v_2 \quad (12)$$

Solving this system of equations, we yield

$$A_v = \frac{v_o}{v_i} \approx -0.1436 \text{ V/V}$$

By superposition, we obtain

$$\begin{aligned} v_o(t) &= V_C^{DC} + v_C^{AC}(t) \\ &= V_C^{DC} + A_v v_i(t) \\ &= 3.449 - 1.436 \sin(2t) \text{ V} \end{aligned}$$

In summary, there are some important equations for analyzing diodes and BJTs with small signals

Diode

Shockley Diode Equation

$$I_D = I_s (e^{V_D/nV_T} - 1)$$

Dynamic Resistance

$$r_d = \frac{dV_D}{dI_D} = \frac{nV_T}{I_{DQ}}$$

Bipolar Junction Transistor (BJT)

Collector Current

$$I_C = I_S e^{V_{BE}/V_T}$$

Transconductance

$$g_m = \frac{dI_C}{dV_{BE}} = \frac{I_{CQ}}{V_T}$$

Hybrid- π resistance

$$r_\pi = \frac{dV_{BE}}{dI_B} = \frac{\beta}{g_m}$$

3. Conclusion

At first glance, this may seem surprising since superposition is strictly valid only for linear systems, whereas devices such as diodes or BJTs are inherently nonlinear. The key idea is linearization.

By first determining the quiescent (DC operating) point of a nonlinear device, we can approximate its behavior near that operating point with a linear small-signal model. Once the circuit has been linearized, the principles of linear circuit analysis, including superposition, become applicable. This allows electrical engineers to separate the DC bias conditions from the AC signal variations and analyze their effects independently.